

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1419

Divide 138 kV
January 23, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1419. The Interconnection Customer proposes a 104.2 MW wind facility interconnecting at the Divide 138 kV substation in the Pennsylvania area. The request was received on March 05, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 551 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,455,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
PINEY CRK PROJECT 345-138 kV XFMR	PINEY CRK PROJECT 500/345kV Autotransformer	111.1	625	46.1
ALLEGHENY HYDRO NO 8 345 kV/138 kV XFMR	PINEY CRK PROJECT / CALDWELL 345kV	111.0	564	13.7
CORYDN / JUANA DIAZ 5802 138kV	PINEY CREEK PROJECT SUBSTATION / CALDWELL 345kV	109.6	202	5.4
PINEY CREEK PROJECT / GEYSERS TAP 1 138kV	PINEY / PINEY CREEK PROJECT 138kV	109.1	413	57.7
RAUN / GEYSERS TAP 1 SUBSTATION 138kV	PINEY CRK PROJECT / PINEY 138kV	109.0	368	72.0
RAUN / DIVIDE 138kV	PINEY / PINEY CREEK PROJECT 138kV	108.8	372	72.0
SCOTT / AMERICAN CRYSTAL SUGAR DRAYTON 345kV	ORCHARD ROAD / SHAWVILLE STATION 500kV	101.0	1,119	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PINEY CRK PROJECT 345-138 kV XFMR	Replace transformer with higher-rated unit	\$19,885,000	10% (\$1,905,000)
ALLEGHENY HYDRO NO 8 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$15,181,000	7% (\$1,069,000)
CORYDON / JUANA DIAZ 5802 138kV	Reconductor line	\$7,314,000	6% (\$441,000)
PINEY CREEK PROJECT / GEYSERS TAP 1 138kV	Reconductor line	\$1,183,000	13% (\$159,000)
RAUN / GEYSERS TAP 1 SUBSTATION 138kV	Reconductor line	\$1,097,000	18% (\$195,000)
RAUN / DIVIDE 138kV line	Reconductor line	\$3,181,000	18% (\$565,000)
AMERICAN CRYSTAL SUGAR DRAYTON / SCOTT 345kV	Reconductor line	\$24,640,000	9% (\$2,121,000)
	Total allocated to project		\$6,455,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.